torch.linspace#

https://pytorch.org/docs/stable/generated/torch.linspace.html

Description:

Creates a one-dimensional tensor of size steps whose values are evenly spaced from start to end, inclusive. That is, the value are: From PyTorch 1.11 linspace requires the steps argument. Use steps=100 to restore the previous behavior. start (float or Tensor) – the starting value for the set of points. If Tensor, it must be 0-dimensional end (float or Tensor) – the ending value for the set of points. If Tensor, it must be 0-dimensional steps (int) – size of the constructed tensor

Function Signature

```
torch.linspace(start, end, steps, *, out=None, dtype=None,
layout=torch.strided, device=None, requires_grad=False) →
Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – the output tensor. dtype (torch.dtype, optional) – the data type to perform the computation in. Default: if None, uses the global default dtype (see torch.get_default_dtype()) when both start and end are real, and corresponding complex dtype when either is complex. layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

Code Examples

```
>>> torch.linspace(3, 10, steps=5)
tensor([  3.0000,   4.7500,   6.5000,   8.2500,  10.0000])
>>> torch.linspace(-10, 10, steps=5)
tensor([-10.,  -5.,   0.,   5.,  10.])
>>> torch.linspace(start=-10, end=10, steps=5)
tensor([-10.,  -5.,   0.,   5.,  10.])
>>> torch.linspace(start=-10, end=10, steps=1)
tensor([-10.])
```

Detailed Documentation

Documentation

Creates a one-dimensional tensor of size steps whose values are evenly spaced from start to end, inclusive. That is, the value are:

From PyTorch 1.11 linspace requires the steps argument. Use steps=100 to restore the previous behavior.

start (float or Tensor) – the starting value for the set of points. If Tensor, it must be 0-dimensional

end (float or Tensor) – the ending value for the set of points. If Tensor, it must be 0-dimensional

steps (int) – size of the constructed tensor

out (Tensor, optional) – the output tensor.

dtype (torch.dtype, optional) – the data type to perform the computation in. Default: if None, uses the global default dtype (see torch.get_default_dtype()) when both start and end are real, and corresponding complex dtype when either is complex.

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.
